

Passaggio parametri versione n. 1

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int q,[IN x copia] int t)
    int r;
      procedure p3([IN OUT x copia] int q)
        int t;
        BEGIN
          t=s;
          if s>50 then q=r else q=s;
          s=2;
          r=s*4;
          p4(t);
          write(q,r,s,t);
        END

      procedure p4([IN OUT x copia] int r)
        int q;
        BEGIN
          q=r;
          r=3;
          t=1;
          s=1;
          write(q,r,s,t);
        END

      BEGIN
        r=4;
        q=t;
        t=r+4;
        s=q;
        p3(r);
        write(q,r,s,t);
      END

  BEGIN
    q=3;
    r=3;
    s=2;
    t=1;
    p2(s, t);
    write(q,r,s,t);
  END
```

Passaggio parametri versione n. 2

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([MODE] int c)
    int a; int d;
      procedure p3([IN OUT x copia] int c,[IN OUT x copia] int d)
        int a;
        BEGIN
          a=3;
          c=d-4;
          d=c;
          b=b*1;
          p4(c, d);
          write(a,b,c,d);
        END

      BEGIN
        a=3;
        d=1;
        c=4;
        b=d*3;
        p3(b, d);
        write(a,b,c,d);
      END

    procedure p4([IN OUT x rif] int a,[IN OUT x rif] int c)
      int b;
      BEGIN
        b=a;
        a=3;
        c=b+1;
        d=c*4;
        write(a,b,c,d);
      END

    BEGIN
      a=4;
      b=4;
      c=1;
      d=2;
      p2(d);
      write(a,b,c,d);
    END
```

Passaggio parametri versione n. 3

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN OUT per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([IN x copia] int t,[MODE] int z)
    int y;
      procedure p3([IN OUT x rif] int z)
        int y; int t;
        BEGIN
          y=z-3;
          t=x+4;
          if z>5 then z=y else z=2;
          x=1;
          write(x,y,z,t);
        END

      BEGIN
        y=4;
        t=3;
        if t>2 then z=4 else z=x-1;
        x=3;
        p3(z);
        write(x,y,z,t);
      END

    procedure p4([IN OUT x rif] int y)
      int t; int x;
      BEGIN
        t=y-2;
        x=z;
        y=y;
        z=x;
        p2(x, z);
        write(x,y,z,t);
      END

    BEGIN
      x=4;
      y=0;
      z=4;
      t=4;
      p4(y);
      write(x,y,z,t);
    END
```

Passaggio parametri versione n. 4

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per riferimento

```
program p1
int x; int y; int z; int t;
  procedure p2([IN OUT x rif] int t,[MODE] int z)
    int y;
      procedure p3([IN OUT x copia] int y)
        int z; int t;
        BEGIN
          z=y;
          t=x;
          y=x-3;
          x=3;
          p4(z);
          write(x,y,z,t);
        END

      procedure p4([IN x copia] int y)
        int x; int z;
        BEGIN
          if t<5 then x=3 else x=4;
          z=4;
          y=x*1;
          t=z;
          write(x,y,z,t);
        END

      BEGIN
        y=3;
        t=2;
        z=x+4;
        x=2;
        p3(z);
        write(x,y,z,t);
      END

  BEGIN
    x=0;
    y=4;
    z=3;
    t=1;
    p2(z, y);
    write(x,y,z,t);
  END
```

Passaggio parametri versione n. 5

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([MODE] int x,[IN OUT x rif] int t)
    int y;
    BEGIN
      y=2;
      if z=7 then x=z else x=y;
      t=x;
      z=4;
      p3(y);
      write(x,y,z,t);
    END

    procedure p3([IN OUT x rif] int x)
      int z; int y;
      procedure p4([IN x copia] int y)
        int z; int x;
        BEGIN
          z=4;
          x=1;
          y=2;
          t=y;
          write(x,y,z,t);
        END

        BEGIN
          z=4;
          y=1;
          if z>9 then x=z-2 else x=2;
          t=t-1;
          p4(x);
          write(x,y,z,t);
        END

      BEGIN
        x=4;
        y=0;
        z=1;
        t=0;
        p2(t, x);
        write(x,y,z,t);
      END
```

Passaggio parametri versione n. 6

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN OUT per riferimento

```
program p1
int x; int y; int z; int t;
  procedure p2([IN OUT x copia] int t,[IN x copia] int y)
    int x;
      procedure p3([IN OUT x copia] int y,[IN x copia] int z)
        int t;
        BEGIN
          t=1;
          y=3;
          z=4;
          if t<9 then x=4 else x=3;
          write(x,y,z,t);
        END

      procedure p4([MODE] int t,[IN x copia] int x)
        int z;
        BEGIN
          z=4;
          t=3;
          x=1;
          y=x;
          p3(z, y);
          write(x,y,z,t);
        END

    BEGIN
      if z>2 then x=3 else x=1;
      t=z*4;
      if y=3 then y=1 else y=1;
      z=3;
      p4(t, x);
      write(x,y,z,t);
    END

  BEGIN
    x=1;
    y=4;
    z=3;
    t=4;
    p2(t, y);
    write(x,y,z,t);
  END
```

Passaggio parametri versione n. 7

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([IN OUT x rif] int y)
    int x; int t;
    BEGIN
      x=z+4;
      t=2;
      if t>3 then y=3 else y=2;
      z=2;
      write(x,y,z,t);
    END

    procedure p3([IN OUT x copia] int z)
      int y; int t;
      procedure p4([MODE] int x,[IN x copia] int z)
        int y;
        BEGIN
          y=2;
          x=4;
          z=2;
          t=2;
          p2(z);
          write(x,y,z,t);
        END

        BEGIN
          y=z;
          t=x;
          z=2;
          x=1;
          p4(z, t);
          write(x,y,z,t);
        END

      BEGIN
        x=1;
        y=3;
        z=0;
        t=2;
        p3(x);
        write(x,y,z,t);
      END
```

Passaggio parametri versione n. 8

- Scoping dinamico, MODE = IN per copia
- Scoping statico, MODE = OUT per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([MODE] int z)
    int t; int x;
    BEGIN
      t=z;
      x=2;
      z=3;
      y=y-3;
      p3(x);
      write(x,y,z,t);
    END

    procedure p3([IN OUT x rif] int x)
      int y; int z;
      procedure p4([IN OUT x copia] int t)
        int x;
        BEGIN
          x=z;
          t=x+2;
          z=3;
          y=3;
          write(x,y,z,t);
        END

        BEGIN
          y=1;
          z=t;
          x=3;
          t=3;
          p4(x);
          write(x,y,z,t);
        END

      BEGIN
        x=0;
        y=0;
        z=3;
        t=1;
        p2(x);
        write(x,y,z,t);
      END
```

Passaggio parametri versione n. 9

- Scoping dinamico, MODE = IN OUT per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN OUT x copia] int t)
    int q;
    BEGIN
      q=s*1;
      t=4;
      r=s+3;
      if q=7 then s=4 else s=s+1;
      write(q,r,s,t);
    END

    procedure p3([IN OUT x copia] int s)
      int q; int r;
      BEGIN
        q=t+1;
        r=t+1;
        s=4;
        t=2;
        p4(q);
        write(q,r,s,t);
      END

      procedure p4([MODE] int s)
        int q; int r;
        BEGIN
          q=2;
          r=s*1;
          s=2;
          t=4;
          p2(s);
          write(q,r,s,t);
        END

      BEGIN
        q=0;
        r=2;
        s=4;
        t=0;
        p3(s);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 10

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([MODE] int d)
    int a; int b;
      procedure p3([IN OUT x rif] int a)
        int c; int b;
        BEGIN
          if a>9 then c=4 else c=3;
          b=a;
          a=a*1;
          d=a-3;
          write(a,b,c,d);
        END

      procedure p4([IN OUT x copia] int a,[IN OUT x copia] int d)
        int c;
        BEGIN
          c=b*1;
          a=b*3;
          d=c;
          b=d*1;
          p3(b);
          write(a,b,c,d);
        END

      BEGIN
        a=c-1;
        b=2;
        d=2;
        c=1;
        p4(c, b);
        write(a,b,c,d);
      END

  BEGIN
    a=3;
    b=2;
    c=3;
    d=2;
    p2(c);
    write(a,b,c,d);
  END
```